



ST BENEDICT'S

SUBJECT

**Physical
Science Paper 2**

DATE

5 August 2019

GRADE

11

MARKS

162

EXAMINER

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MODERATOR

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NAME

DURATION

150 minutes

CLASS

**LOW
ORDER**

55%

HIGH ORDER

45%

QUESTION 1- JOOSTE

18 MARKS

- 1.1 B✓✓
- 1.2 A✓✓
- 1.3 B✓✓
- 1.4 D✓✓
- 1.5 A✓✓
- 1.6 D✓✓
- 1.7 C✓✓
- 1.8 B✓✓
- 1.9 B✓✓

QUESTION 2 RAMS

23 MARKS

2.1

- 2.1.1 A bond which occurs between atoms within molecules ✓✓ (AON) (2)
- 2.1.2 Polar ✓ covalent ✓ bonds (2)
- 2.1.3 Gas ✓ (1)
- 2.1.4 Liquid/Molten. ✓ (1)
- 2.1.5 CO₂ (exists in a molecular lattice where) **molecules** ✓ are held together by weak **London forces** ✓ (or intermolecular forces)
- SiO₂ (exists in an atomic lattice where) **atoms** ✓ are held together by strong (polar) **covalent bonds** ✓
- More energy is required to **break strong bonds** than to **overcome weak forces** ✓ (higher melting point) (5)

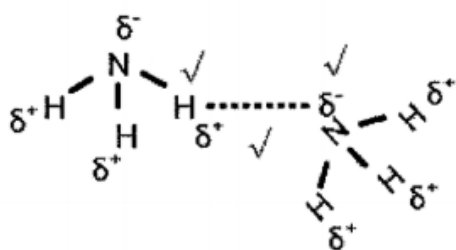
2.2

- 2.2.1 A weak force of attraction between discrete molecules, ions or atoms of noble gases. ✓✓(AON) (2)
- 2.2.2 Dipole-dipole forces. ✓✓ (2)
- 2.2.3 $n = c \times v$ ✓ (Formula)
- 0,2 x 0,25 ✓ (Subs)
- = 0,05 mol ✓
- $m = n \times M$
- $m = 0,05 \times (90 + 36)$ ✓ (Subs)
- $m = 6,3 \text{ g}$ ✓ (5)
- 2.2.4 Solute must be added first ✓. It will not have correct concentration ✓. The volume of the solution will be greater, as the particles will add to the total volume. ✓ (3)

QUESTION 3 JOOSTE

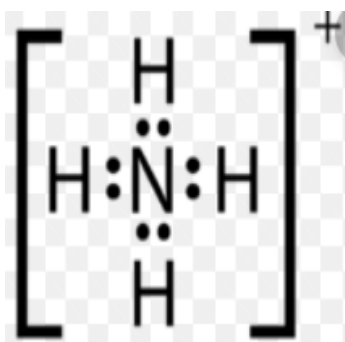
24 MARKS

- 3.1 CH₄ contains polar bonds ✓
But the molecule has electron✓ symmetry ✓
So it is non-polar✓ (4)
- 3.2 IMF in NH₃ is hydrogen bond force, very strong IMF✓, require a high amount of energy to overcome IMF✓, ∴ high Bpt✓. CH₄ is London forces✓ (4)
- 3.3 PH₃ a higher Boiling point than CH₄✓
Therefore PH₃ has a stronger intermolecular force✓ (2)
- 3.4.1 High electronegativity, small atom OR At least one lone pair.(Any 2) (2)
- 3.4.2 (3)



- ✓ Couper structure correct
- ✓ Slightly negative on N and slightly positive on H (don't have to show on all)
- ✓ Interaction shown (H-N)

- 3.4.3 Sigma bonds✓ (1)
- 3.4.4 A bond involving a sharing of at least one electron pair✓, by 2 non metal atoms✓ (2)
- 3.4.5 When a lone pair of an atom overlaps✓ the empty orbital of another atom ✓ (2)
- 3.4.6 (3)



- ✓ ion charge
- ✓ all electrons correct
- ✓ 4 hydrogens and one nitrogen

QUESTION 4 MCCOY

28 MARKS

4.1.1 $n = m/M$ (✓formula)

$n = 500$ (✓molar mass)

$n = 0,05 \text{ mol}$

(2)

4.1.2

$n \text{ of HCl} = c \cdot V$ (formula)

$n = 2 \times 0,5$ (✓conversion of volume)

$n = 0,25 \text{ mol}$ (✓answer)

$\text{CaCO}_3 + 2\text{HCl}^*$

Option A: 0,05 0,1

Option B: 0,25 0,5

Option A is valid

(✓method)

Therefore CaCO_3 is the limiting reagent (✓answer)

(4)

***coe**

4.1.3 $n_{(\text{HCl})} = 0,5 - 0,1$

$= 0,4 \text{ mol}$

$c = n/v$ (formula)

$v = n/c$

$0,4 \checkmark / 2 \checkmark$

$= 0,2 \text{ dm}^3 \checkmark$

(3)

4.2.1 A Temporary transition state between the reactants and the products. ✓✓

4.2.2

I = energy of reactants $E_{\text{reactants}}$ / Energy before reaction / Enthalpy of reactants
II = activation energy / E_A / Energy to break bonds
III = energy released (on product formation) E_{released} / bonding energy
IV = energy of products / E_{products}
V = heat of reaction / change in enthalpy / ΔH / enthalpy / change in **net** potential energy (5)

NOT ACCEPTING – I = initial energy / energy level

III = energy lost

V = change in heat of reaction

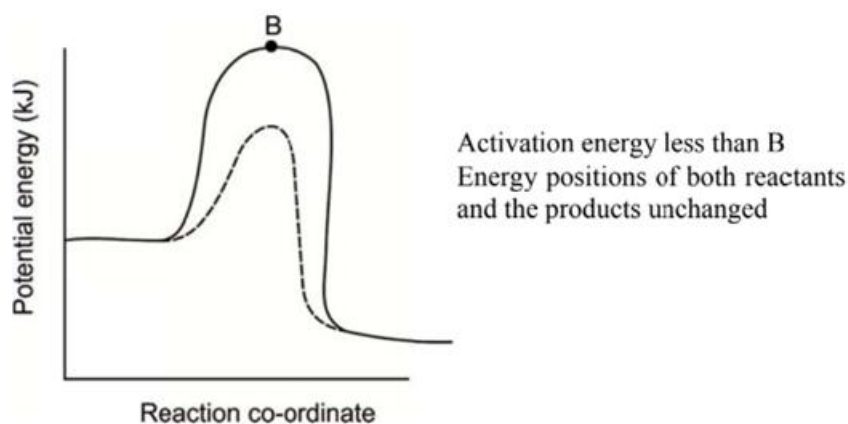
4.2.3

- Exothermic reaction
 - The amount of energy released on product formation is greater than the amount of energy absorbed to reach the activated complex. (3)
- OR**
- Exothermic reaction
 - The amount of energy of the products is less than the amount of energy of the reactants.
- OR**
- Exothermic reaction
 - The heat (enthalpy) of the reaction will be negative / ($\Delta H < 0$)

If they say Endothermic reaction – but with clear explanation (2/3) otherwise (0/3)

4.3.1 A substances that increases the rate of reaction but remains unchanged by the end of the reaction. ✓✓

4.3.2



(2)

4.3.3

- No change ✓
- Catalyst only lowers the energy of activation ✓ and does not affect the final energy of the products ✓ formed.

(3)

OR

Does not change the energy of reactants and products

OR

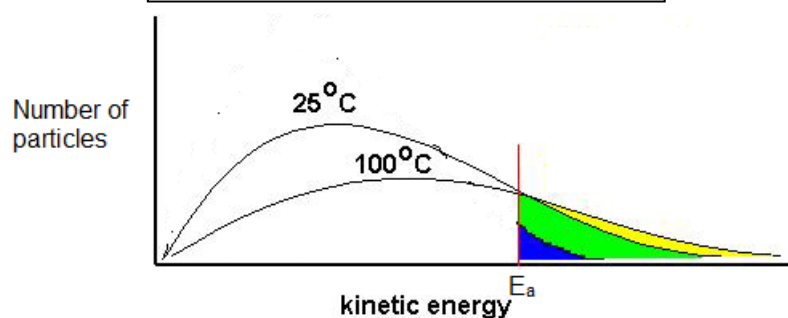
I and IV remain constant

4.3.4

(2)

✓ 10 degrees' curve below 25 degrees' curve.

✓ Same activation energy



QUESTION 5 NTOMBELA**31 MARKS**

- 5.1 The change in concentration per unit time ✓ of either a reactant or product ✓ (2)
- 5.2 Some mass ✓ is released as a gas ✓ (2)
- 5.3 Concentration ✓ of HCl ✓ (2)
- 5.4 Temperature ✓ OR volume of HCl ✓ (2)

5.5

- 5.5.1 The net change of chemical potential energy of the system. ✓✓ (AON) (2)

- 5.5.2 The reaction is exothermic ✓ (2)

Resulting in the temperature initially increasing (releases heat) ✓

- 5.5.3 As time proceeds the concentration of HCl decreases ✓ (4)

Fewer particles per unit volume resulting in fewer collisions per second. ✓

This results in fewer effective collisions per second (sufficient kinetic energy and correct orientation). ✓

Leading to a decreasing reaction rate. ✓

- 5.5.4 (5)

$$\begin{aligned}n(\text{Zn}) &= \frac{m}{M} \\&= \frac{0,5}{65,4} \\&= 7,65 \times 10^{-3} \text{ mol } \checkmark\end{aligned}$$

Ratio 1:1

$$\begin{aligned}n(\text{H}_2) &= 7,65 \times 10^{-3} \text{ mol } \checkmark \\&\text{ratio}\end{aligned}$$

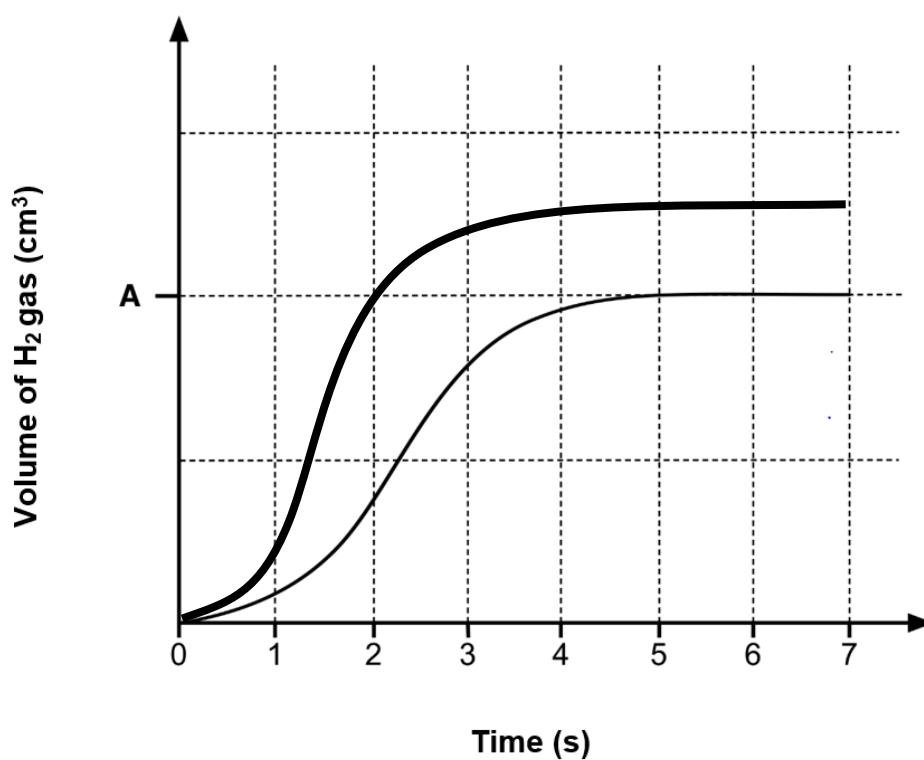
$$\begin{aligned}V &= nV_m \checkmark \text{ formula} \\&= (7,65 \times 10^{-3})(22,4) \\&= 0,171 \text{ dm}^{-3} \checkmark \text{ answer} \\&= 171,25 \text{ cm}^3 \checkmark \text{ conversion}(\text{coe})\end{aligned}$$

5.5.5 % Accuracy = $\frac{\text{Actual} - \text{Experimented}}{\text{Actual}} \times 100$ ✓

$\frac{200 - 171,225(\text{coe})}{200} \times 100$ ✓

14,38 % ✓

5.5.6



✓ Steeper and ends sooner

✓ Ends above A

(2)

5.6 No ✓

The state of the Zn was not controlled ✓

Surface area also affects rate ✓

(4)

Experiment not repeated more than 3 times ✓

QUESTION 6 DUTOIT

18 MARKS

6.1.1 Molar mass is defined as the mass (in grams) of one mole of a particular substance ✓✓ (all or nothing) [2]

6.1.2 $M = (7 \times 12) + 6 + (3 \times 16)$ ✓ = 138 g/mol ✓(unit required) [2]

6.1.3 $n = m / M$ ✓ = $3.86 / 138$ (coe)✓ = 0.028 moles ✓
1 salicylic acid : 1 aspirin therefore 0.028 moles aspirin produced✓
 $m = n \times M = 0.028 \times (180)$ ✓ = 5.04 g✓ [6]

6.1.4 % yield = 4.65 g ✓ / 5.03 g (coe)✓ * 100 = 92.26%✓ [3]

6.2 $n = m / M = (300 \times 10^{-3}) / 180$ (coe)✓ = 0.00167 moles✓
1 mole aspirin = 8 mole hydrogens ✓
Therefore $0.00167 \times 8 = 0.0133$ moles H atoms
 $0.0133 \times (6.02 \times 10^{23})$ ✓ = 8.03×10^{21} atoms of hydrogen ✓ [5]

QUESTION 7 DUTOIT**20 MARKS**

7.1.1 The substance in which another substance is dissolved, forming a solution. ✓✓ (2)

7.1.2 Water ✓ (1)

7.1.3 $n = c \times v$ ✓(formula)

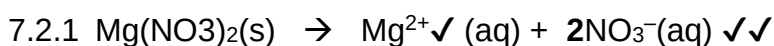
$$n = 1.5 \times 0,5 \text{ ✓ (subs)}$$

$$n = 0,75 \text{ mol}$$

$$0,25 = 0,75/v \text{ ✓(subs – moles remain unchanged)}$$

$$V = 3 \text{ dm}^3 \text{ ✓}$$

$$\text{Volume of water added} = 3 - 0,5 = 2,5 \text{ dm}^3 \text{ ✓} \quad (5)$$



Phase indicators all correct ✓ (4)

7.2.2 $n = c \cdot V$

$$= 0,15 \times 0,2 \text{ (✓conversion)}$$

$$= 0,03 \text{ mol} \times 2 \text{ (✓multiplication by 2) (coe)}$$

$$= 0,06 \text{ mol (✓answer)} \quad (3)$$

$$7.2.3 \quad x + 0,06 \times 0,2 = 0,50 \text{ ✓✓}$$

$$x = 0,04 \text{ mol ✓}$$

OR

$$\begin{aligned} n &= c \cdot V \\ &= 0,5 \times 0,2 \\ &= 0,1 \text{ mol of } \text{NO}_3^- \text{ ions in total} \end{aligned}$$

$$\begin{aligned} n &= (0,1 - 0,06) = 0,04 \text{ mol of } \text{NO}_3^- \text{ ions added} \\ &\text{(0,06 c.o.e from Question 7.2.2)} \end{aligned}$$

$$\begin{aligned} m &= n \times M \\ &= 0,04 \times 85 \\ &= 3,4 \text{ g of NaNO}_3 \end{aligned}$$

✓✓✓

$$n = m/M$$

$$0,04 = m/85 \text{ ✓}$$

$$m = 3,4 \text{ g ✓}$$

(5)

END OF EXAMINATION